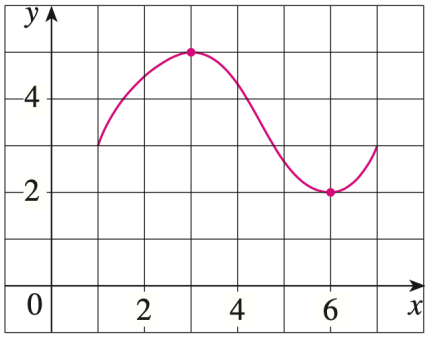
Some of the most important applications of differential calculus are *optimization problems*, in which we are required to find the optimal (best) way of doing something. Such problems can often be reduced to finding the maximum or minimum values of a function.

Definition. Let c be a number in the domain D of a function f. Then $f(c)$ is the

- *absolute maximum* value of f on D if $f(c) \geq f(x)$ for all x in D.
- *absolute minimum* value of f on D if $f(c) \leq f(x)$ for all x in D.

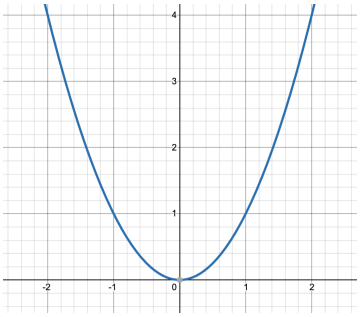
An absolute maximum or minimum is also called a *global* maximum or minimum. An *absolute extreme value* is either an absolute maximum or an absolute minimum.

1.) Consider the graph of a function f below. Find the absolute extreme values of f.

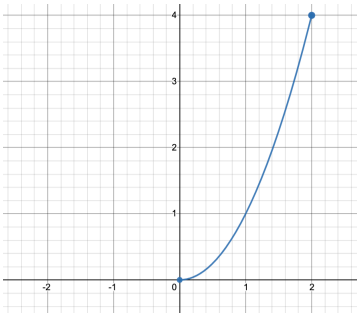


The existence and location of absolute extreme values depend on both the function and the interval of interest.

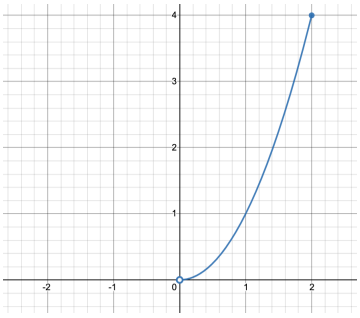
2.) Consider the function $f(x) = x^2$. On each interval below, determine the absolute max and absolute min, if these exist.



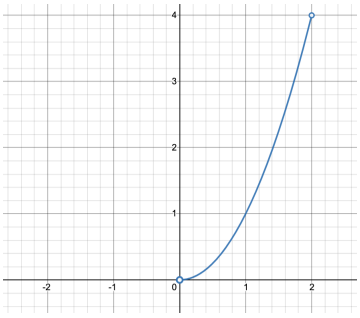
(a) $f(x) = x^2$ on $(-\infty, \infty)$



(b) $f(x) = x^2$ on $[0,2]$



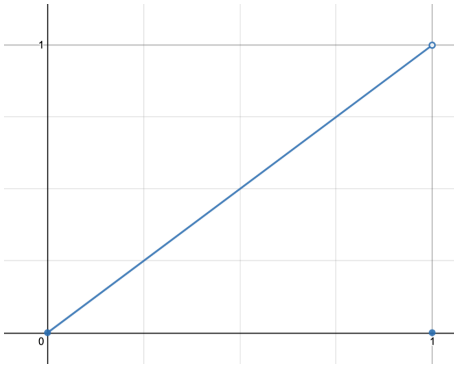
(a) $f(x) = x^2$ on $(0,2]$

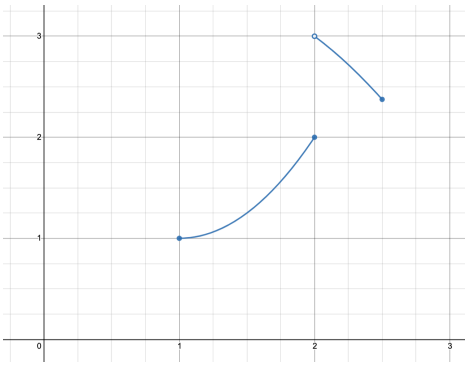


(b) $f(x) = x^2$ on $(0,2)$

3.) Based on your observations above, what must be true about the interval for a function to have an absolute maximum and an absolute minimum?

4.) Both functions below are defined on closed intervals. What must be true about a function for it to have an absolute maximum and an absolute minimum?

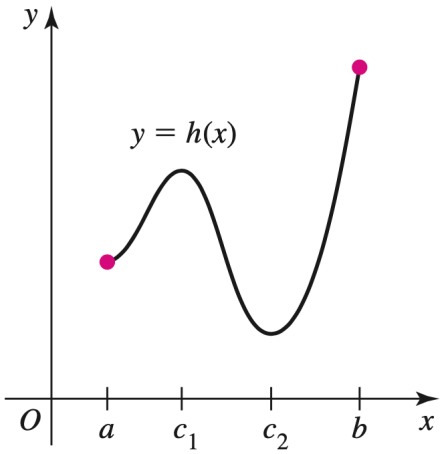


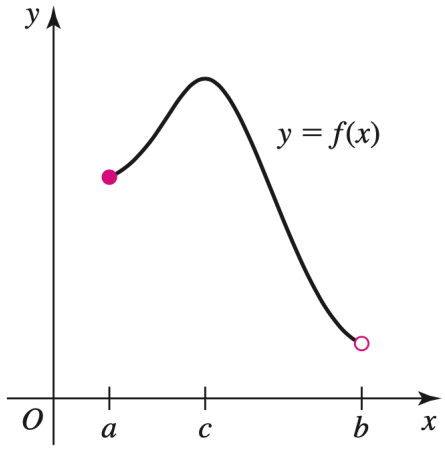


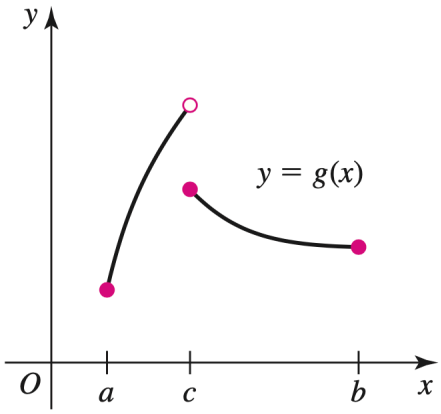
Name:_____

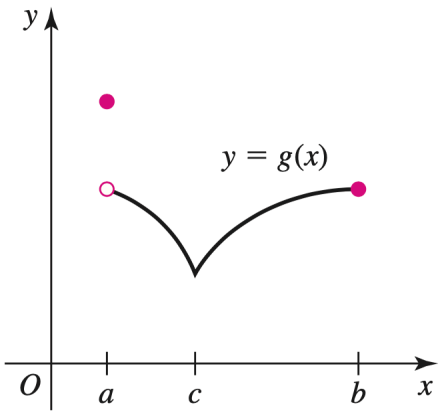
Extreme Value Theorem A function that is continuous on a closed interval $[a,b]$ has an absolute maximum and an absolute minimum on that interval.

5.) For each graph below, determine whether (i) the hypotheses and / or (ii) the conclusions of the Extreme Value Theorem are satisfied on the interval $[a,b]$.

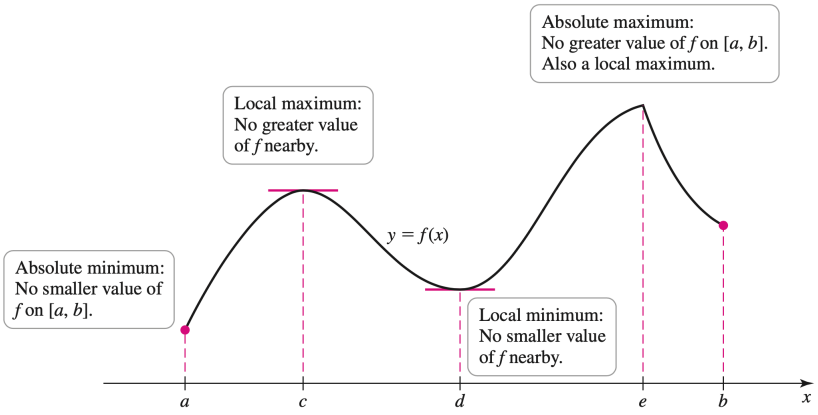








The figure below shows a function defined on the interval $[a,b]$. The function has an absolute minimum at the endpoint a and an absolute maximum at the interior point e. In addition, the function has special behavior at c, where its value is the greatest *among values at nearby points*, and at d, where its value is least *among values at nearby points*.



Definition. Let f be defined on an open interval (a,b) containing c.

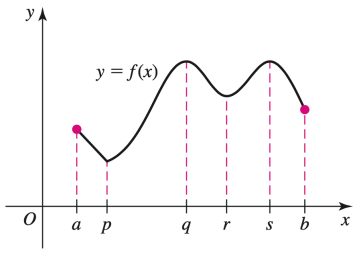
- If $f(c) \geq f(x)$ for all x in (a,b), then $f(c)$ is a *local maximum* value of f.
- If $f(c) \leq f(x)$ for all x in (a,b), then $f(c)$ is a *local minimum* value of f.

A left or right endpoint of the domain of f could also be local maximum or minimum, but we need to alter the definition above since there is no open interval containing an endpoint: Suppose f is defined on a half-open interval $[c,d)$.

- If there exists a half-open interval $[c,b)$ with $b \leq d$ such that $f(c) \geq f(x)$ for all $x \in [c,b)$, then $f(c)$ is a *local maximum* value of f.
- If $f(c) \leq f(x)$ for all $x \in [c,b)$, then $f(c)$ is a *local minimum* value of f.

Similar definitions hold for a right endpoint.

6.) Locate all local and absolute extrema of the function graphed below.



7.) Suppose f is has a local maximum or minimum at a point c and is differentiable at c. What is the slope of the tangent line to the graph of f at $(c, f(c))$?

Theorem. (Fermat's Theorem) If f is has a local maximum or minimum at a point c and is differentiable at c, then $f'(c) = 0$.

(The rest of this page is optional.) The proof makes use of the following fact about limits:

Lemma. If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist at x approaches a, then

$$\lim_{x\to a} f(x) \leq \lim_{x\to a} g(x).$$

Proof of Fermat's Theorem. Suppose f has a local maximum at c (the proof for a local minimum at c is similar). Then, by definition of a local maximum, $f(c) \geq f(x)$ for all x sufficiently close to c. This implies that if h is sufficiently close to 0, with h being positive or negative, then $f(c) \geq f(c+h)$ and therefore $f(c+h) - f(c) \leq 0$. We can divide both sides of an inequality by a positive number without changing the direction of the inequality. Thus, if $h > 0$ and h is sufficiently small, then

$$\frac{f(c+h) - f(c)}{h} \leq 0$$

and by the Lemma

$$\lim_{h\to 0^+} \frac{f(c+h) - f(c)}{h} \leq \lim_{h\to 0^+} 0 = 0.$$

Since $f'(c)$ exists, we therefore have

$$f'(c) = \lim_{h\to 0} \frac{f(c+h) - f(c)}{h} = \lim_{h\to 0^+} \frac{f(c+h) - f(c)}{h} \leq 0.$$

8.) Now consider the case when $h < 0$ and finish the proof.

9.) Is the converse of Fermat's Theorem true? That is, if $f'(c) = 0$, does f necessarily have a local maximum or minimum? Explain. (*Hint*: Consider $f(x) = x^3$.)

10.) Suppose the hypotheses of Fermat's theorem are false. *Could* f still have a local maximum or minimum? Explain. (*Hint*: Consider $f(x) = |x|$.)

Fermat's Theorem tells us that extreme values of f must occur at points c where $f'(c) = 0$ or where $f'(c)$ does not exist.

Definition. A *critical point* of f is a number c in the domain of f where either $f'(c) = 0$ or $f'(c)$ does not exist.

Fermat's Theorem can therefore be restated as follows:

Theorem. (Fermat's Theorem) If f has a local maximum or minimum at c, then c is a critical point of f.

Remark. Note that, if f is defined on a closed interval, then the endpoints are automatically critical points since f is not differentiable at the endpoints (f may have a one-sided derivative at each endpoint, but never a two-sided derivative).

11.) Find the critical points of $f(x) = x^{3/5}(4-x)$.

The previous discussion proves the following theorem:

Theorem. To find the absolute maximum and minimum values of a continuous function f on a closed interval $[a,b]$:

1. Find the values of f at the critical points of f.
2. Evaluate f at each critical point. The largest of these values is the absolute maximum value; the smallest of these values is the absolute minimum value.

12.) Find the absolute maximum and minimum values of the following functions.

(a) $f(x) = x^4 - 2x^3$ on the interval $[-2,2]$

(b) $g(x) = x^{2/3}(2-x)$ on the interval $[-1,2]$.